

4-LAYER DIAGNOSTIC-AWARE TELE-RADIOLOGY ARCHITECTURE (DARDO + CSA + 5G URLLC)

MOBILE STROKE UNIT (AMBULANCE)

LAYER 1: PERCEPTION & AUDIT (CSA)

- Counterfactual: $\mathbf{Y}_{cf} = \mathbf{Y} + \mathbf{A}\delta$
- Gabor Channelised Hotelling Observer (CHO)
- Clinical Detectability Metric ($d' \geq 1.5$)
- *Replaces blind pixel-averaged PSNR/SSIM*

LAYER 2: COMPRESSION (DARDO)

- Task Spectral Density: $W(k_x, k_y) = |\mathcal{F}\{\mathbf{w}\}|^2$
- Diagnostic Gradient: $\gamma_k = \sum_y W(k, y) \|\mathcal{FS}_k\|_2$
- Greedy Line Allocation (< 2 ms on Edge CPU)
- **Cuts silent stroke erasure: 28.8% to <4.5%**

5G/B5G CELLULAR RADIO ACCESS

LAYER 3: 5G URLLC SLICE (CRITICAL)

- High Diagnostic Gradient Bands ($\gamma_k \geq \tau$)
- **Tagged: 5G QCI 1 / 65 (URLLC Slice)**
- Block Error Rate (BLER) $< 10^{-5}$
- *Sub-10 ms latency; immune to fading drops*

LAYER 3: 5G eMBB SLICE (BULK)

- Low Diagnostic Gradient Bands ($\gamma_k < \tau$)
- Best-Effort Enhanced Mobile Broadband
- Carries gross anatomical background
- *Tolerates packet loss without diagnostic harm*

HOSPITAL EDGE CLOUD SERVER

EDGE AI RECONSTRUCTION

- Multi-Coil Deep Variational Net (VarNet)
- Inference Latency: 1.8 s on GPU server
- Recovers patient cross-sectional slice $\hat{\mathbf{X}}$
- *Generates high cosmetic PSNR (> 25 dB)*

LAYER 4: SEMANTIC AUDIT & S-NACK

- 128-Byte Semantic Token in QUIC Header
- On-Edge CHO Audit: score z_{recon} (< 15 ms)
- **Pass: $z \geq 1.5 \Rightarrow$ Verified; Fail: $z < 1.5 \Rightarrow$ Erased!**
- *Targeted S-NACK requests missing Δk lines*

CLOSED-LOOP S-NACK (Requests only missing Δk bands, ~ 15 ms)